

Measuring AI-Washing in Corporate Filings

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Abstract

Investors and regulators face difficulty distinguishing implemented AI capability from narrative positioning in annual 10-K filings. Using U.S. public firms from 2016 to 2025, we classify AI-related sentences in corporate filings as actionable, speculative, or irrelevant and aggregate them into filing-based disclosure measures. We combine low-credibility disclosure composition with contemporaneous AI patent weakness to construct PatentMismatch, a scalable and audited proxy for low-credibility AI disclosure. After 2022, PatentMismatch characterizes more than 40% of AI-talking firm-years in both 2024 and 2025. PatentMismatch predicts weaker subsequent AI patent grants and profitability, and higher subsequent R&D intensity. Disclosure credibility appears stronger following SEC comment-letter scrutiny, weaker around large equity-issuance windows, and lower when CEO equity incentives are stronger. Filing-date reactions remain muted. The evidence points to a disclosure-credibility problem whose strongest patterns appear in later realization, scrutiny, incentives, and financing-window dynamics rather than in immediate market pricing.

Keywords: Artificial intelligence; AI-washing; Corporate disclosure; 10-K filings; Textual analysis; Innovation; Patents

JEL classifications: D82, G14, M41, O31, O32

1 Introduction

Artificial intelligence has become central in corporate disclosure, but outsiders cannot easily verify whether that language reflects implementation. AI-related investments can affect innovation and growth (Babina et al., 2024; Cao et al., 2024). Prior evidence shows that investors treat AI narratives in mandatory filings as informative rather than generic boilerplate (Basnet et al., 2025). In our data, AI language in annual reports expands faster than contemporaneous AI patenting. Figure 1 shows the scale of this gap. To capture it, we build PatentMismatch, an indicator that flags AI-talking firm-years in which the language tilts toward vague rather than concrete AI content and the firm patents little in AI relative to its industry that year. PatentMismatch characterizes more than 40% of AI-talking firm-years in both 2024 and 2025. The post-2022 surge in AI disclosure therefore does not simply mirror a comparable surge in implemented AI capability.

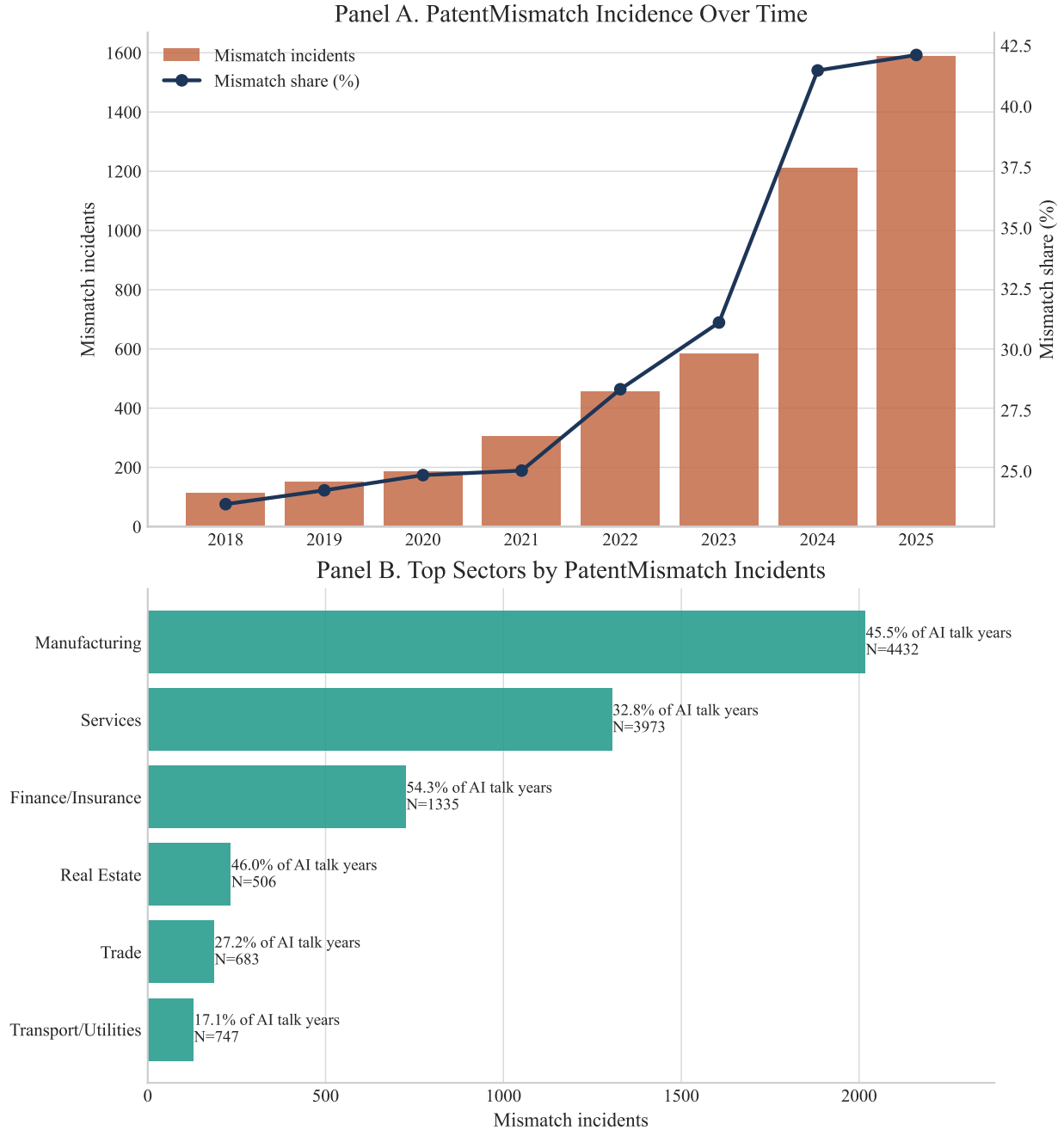


Figure 1: The rise of low-credibility AI disclosure, 2018–2025

Note: This figure plots the share of AI-talking firm-years flagged as PatentMismatch. AI-talking firm-years are firm-years with at least one AI-related sentence in the 10-K filing. Panel A shows the count of flagged firm-years and the mismatch share over time. Panel B shows the mismatch share within the sectors that account for the most AI-talking firm-years.

The credibility problem extends beyond AI. When implementation is costly, salient, and difficult to verify, disclosure can convey information while remaining only loosely tied to realized action. Classic signaling models formalize this logic (Spence, 1973; Seidmann & Winter, 1997). Adjacent washing literatures document the same problem empirically. Public claims can outrun later observable conduct when investors, regulators, or other stakeholders care about the topic (Delmas &

Burbano, 2011; Kim & Lyon, 2015; Lyon & Montgomery, 2015; Baker et al., 2024). AI-related disclosure fits this setting because firms can invoke a valuable technology before outside observers can verify implementation. The central question is whether filing language separates credible AI disclosure from narrative positioning, and whether that distinction predicts realization, organizational response, and market outcomes.

Recent work studies AI communication more directly. The closest published evidence shows that investors respond differently to actionable and speculative AI language in 10-K business descriptions (Basnet et al., 2025). Related work links AI disclosure to later firm outcomes and broader AI narratives to later return patterns (Cao et al., 2024; Bandyopadhyay et al., 2023). In addition, recent working papers compare AI claims with later implementation or capability proxies (Li, 2025; Donelson et al., 2025; Barrios et al., 2025). A remaining gap is an auditable, scalable filing-based framework that links disclosure composition to contemporaneous implementation and later realization.

To fill that gap, we use annual 10-K filings of U.S. public firms from 2016 to 2025. We classify AI-related sentences as actionable, speculative, or irrelevant, aggregate those classifications into firm-year disclosure measures, and use the actionable-to-speculative (AS) ratio together with contemporaneous patent weakness to construct PatentMismatch. The tests first validate PatentMismatch against later AI realization, then examine scrutiny and incentive responses, and finally turn to market outcomes.

The validation evidence begins with later AI realization. PatentMismatch rises sharply after 2022 and becomes common among AI-talking firms. Across alternative construct definitions, the main PatentMismatch measure predicts weaker subsequent AI patent grants. Application-based variants are broader: later AI applications can rise even when near-term patent-grant realization is weak. Among AI-talking firms, mismatch years are followed by lower profitability and higher R&D intensity, consistent with weaker near-term operating performance and later resource reallocation rather than immediate technological realization.

Disclosure credibility also varies with scrutiny, incentives, and financing windows. Among continuing AI disclosers, disclosure composition becomes more credible after first SEC comment-letter scrutiny, without a comparable decline in AI focus, the log-transformed count of AI-related sentences in the annual filing. A public SEC salience event points in the same direction. After the March 18, 2024 AI-washing enforcement actions (SEC, 2024), pre-exposed firms appear to clean up their disclosure composition. The March 2024 pattern is best interpreted as salience or acceleration, not as a sharply identified enforcement effect. Inside the firm, low-credibility AI disclosure is more common where CEO equity incentives are stronger. Around first large equity-issuance windows, credibility deteriorates into the financing event and partly unwinds afterward, while AI focus continues to rise. Low-credibility AI disclosure is concentrated where scrutiny, incentives, and financing conditions should matter.

Return evidence is more limited than the validation and organizational-response evidence. Filing-date abnormal returns do not distinguish mismatch from non-mismatch firms, and the broader market block is mixed and benchmark-sensitive. The benchmarked portfolio evidence is strongest in a short-horizon equal-weight post-filing alpha. The corresponding value-weight evidence is weak. Issue-window tests show that the short-run return relation is state-dependent and less negative inside financing windows than outside them. Return tests therefore point to limited and state-dependent pricing of disclosure credibility rather than a broad pricing-anomaly interpretation.

We contribute a filing-based, auditable construct for low-credibility AI disclosure and evidence on when it matters. PatentMismatch predicts weaker subsequent AI patent grants and weaker operating performance, and it is most common where scrutiny, incentives, and financing windows should bear on narrative credibility. Read together, the results describe AI washing as a disclosure-

credibility problem that links firms’ AI narratives to their later technological and organizational outcomes.

Section 2 places the paper in the disclosure-credibility, washing, and textual-analysis literatures. Section 3 describes the data, the sentence-classification pipeline, the disclosure measures, and the empirical design. Section 4 reports the realization, scrutiny, incentive, financing-window, and market-pricing results. Section 5 discusses what the evidence does and does not establish.

2 Literature Review

The closest conceptual foundation is disclosure credibility. Accounting and finance research does not treat disclosure as a frictionless release of facts; it treats disclosure as communication shaped by information asymmetry, proprietary costs, reputational concerns, and strategic incentives (Healy & Palepu, 2001; Verrecchia, 2001; Beyer et al., 2010). The key implication is more specific. When the activity being described is costly to implement and difficult to verify, the credibility of the language depends on whether lower-quality senders can cheaply mimic higher-quality ones (Spence, 1973; Seidmann & Winter, 1997). AI disclosure fits that environment because filing language can expand more quickly than observable implementation.

The washing literature gives the same credibility problem empirical content. In environmental and sustainability settings, public claims can diverge from realized behavior when firms face strong incentives to appear aligned with valued goals (Delmas & Burbano, 2011; Kim & Lyon, 2015; Lyon & Montgomery, 2015). Marquis et al. (2016) show how selective disclosure can create the appearance of transparency despite weaker underlying records, while Hummel and Schlick (2016) emphasize that disclosure quality matters more than disclosure volume. Baker et al. (2024) extend this logic to diversity disclosure, where claims can outpace later workforce outcomes. They motivate a test of whether valued disclosure aligns with implementation and later realization, beyond whether such language appears.

Recent AI-disclosure papers bring the credibility question closer to the present setting. Basnet et al. (2025) provide the closest published starting point on the language side by distinguishing actionable, speculative, and irrelevant AI disclosure in 10-K business descriptions and showing that investors respond differently across those categories. Bandyopadhyay et al. (2023) are closest on the market side: they show that broader AI narratives are associated with subsequent stock mispricing, especially among firms outside the largest size group. Related working papers study AI disclosure, disclosure–implementation gaps, and overstatement more directly (Li, 2025; Donelson et al., 2025; Barrios et al., 2025). They leave open how disclosure composition, contemporaneous implementation, and later realization can be linked without treating them as the same object. PatentMismatch addresses that gap by using filing-level disclosure composition to identify low-credibility AI narratives and relate them to realization, scrutiny, incentives, and market outcomes.

Textual-analysis research reaches a similar lesson from the measurement side: corporate language is informative only when it is measured in a way that fits the institutional setting. Forward-looking filing language predicts later fundamentals (Li, 2010), earnings-press-release language adds incremental information (Davis et al., 2012), and results are sensitive to the underlying vocabulary design (Loughran & McDonald, 2011, 2016). Tone and wording can also be managed strategically rather than passively reflecting fundamentals (Huang et al., 2014). Narrative disclosure is not unreliable simply because it can be managed. Its empirical content depends on whether the measure isolates the relevant linguistic dimension rather than collapsing disclosure into a broad mention count. The distinction matters in an AI setting, where generic references, aspirational language, and implemented uses can all appear in the same filing while conveying very different information.

The innovation literature points in the same direction. Bellstam et al. (2021) show that richer textual measures capture meaningful differences in firm-level innovation and forecast subsequent patenting and performance. Kim and Valentine (2023) show that public disclosure improves how markets interpret innovation-related assets. Together, these studies motivate a two-sided validation approach. Filing language can contain information about later technological activity, but measurement choices determine how much of that information is recovered. The narrower question is whether disclosure composition isolates filings less aligned with contemporaneous implementation and subsequent technological realization.

3 Data and Methodology

3.1 Sample construction and data sources

We combine annual 10-K filings, AI-related patents, CRSP market data, and Compustat controls. Annual 10-K filings provide the sentence-level corpus for the AI-disclosure measures. The innovation outcomes are constructed from AI-related patent grants drawn from USPTO/PatentsView and matched to public firms through a procedure that links CIK, ticker, GVKEY, and patent-assignee identifiers. Market outcomes are drawn from CRSP and are used for filing-date abnormal-return tests and post-filing return analyses. Firm characteristics and annual controls come from Compustat.

For the filing-text layer, we use the Notre Dame Software Repository for Accounting and Finance post-Stage One 10-X files as the upstream source for annual filing text (McDonald, 2022; Software Repository for Accounting and Finance, 2026). The Stage One parse removes major non-text artifacts from EDGAR filings, including markup, XBRL/XML content, ASCII-encoded segments, tables, and excess linefeeds. We then apply filing-level filters, reconstruct sentences, screen for AI terms, remove duplicates, and classify sentences.

The main annual sample covers 2016–2025 and includes every firm that mentions AI at least once during that period. We begin in 2016 because AI references in 10-K filings are sparse before then, whereas the broader expansion of AI-related disclosure becomes economically visible only later in the sample. Once a firm enters the sample, we retain all of its available annual observations, including years with zero AI-related disclosure, that is, the years in which the firm’s filing contains no AI-related language at all. Keeping these zero-disclosure years lets us line up prior, contemporaneous, and later outcomes on a common annual timeline. The resulting panel contains 50,840 firm-year observations across 5,084 firms.

Within this sample, 13,777 firm-years contain at least one classified AI sentence, and the linked filing-event sample contains 7,355 annual filing events across 2,753 firms, of which 7,342 have usable $CAR[-1, +1]$ returns and 7,098 have usable $BHAR[+2, +63]$ outcomes.

The sample also supports several linked outcome blocks. One block validates the disclosure construct against later AI-related realization, including patent grants and applications. Other blocks study filing-date and post-filing returns, annual valuation and financing outcomes, real outcomes, and organizational responses around scrutiny, incentives, and issuance windows. The market tests use CRSP-linked filing-event samples and monthly return panels, while the innovation tests use the annual firm-year panel. Not every empirical block uses exactly the same observations. Sample sizes change mechanically because some specifications require AI-talking firm-years, some require lead or lag patent outcomes, and some require complete cases on the annual control set. The filing-date event-study tests also impose CRSP availability and event-window requirements. Appendix Table A2 reports the resulting attrition across the main empirical blocks.

The analysis does not rely on a single estimation sample. We construct one common filing-based disclosure layer and then link it to separate annual and event-time outcomes. The common dis-

closure layer lets us ask whether the same disclosure-credibility construct predicts later realization, organizational response, and market outcomes.

We use AI-talking firm-year to refer to any firm-year with at least one classified AI sentence. Event-study tables sometimes impose stronger activity screens; in those cases, active AI disclosers are firms that continue mentioning AI in the relevant pre- and post-event windows, and AI issuers are active AI disclosers around large equity-issuance windows.

3.2 Sentence extraction and AI-language classification

We segment each filing into sentences and remove duplicate blocks, tables, headers, and other clearly non-narrative content. We identify AI-related sentences using a fixed keyword list that we update as new AI terms enter common use. The list includes terms such as “artificial intelligence,” “machine learning,” “deep learning,” “neural network,” “natural language processing,” “computer vision,” “robotic process automation,” “large language model,” and “generative AI.” Each retained sentence is keyed to the firm and fiscal year of the filing.

The core measurement step is a three-way sentence classification. Actionable sentences describe concrete AI deployments, embedded workflows, internal tools, or productized uses. Speculative sentences refer to AI in aspirational, exploratory, or forward-looking terms without enough operational detail to verify current implementation. Irrelevant sentences contain generic or tangential AI references that do not speak to the firm’s own capability or future plans.

The scaling pipeline has three distinct validation layers. First, we build a human labeling rubric and conduct an inter-rater-reliability audit on a stratified validation subset. Second, we evaluate a tuned three-stage classifier against the adjudicated labels and use that model to scale the classification across the full sentence corpus. Third, we validate the resulting firm-year measures against independently constructed AI patent outcomes. Separating these layers keeps rubric reliability, classifier accuracy, and external validation conceptually distinct. In the expanded 2016–2025 sample, the classified corpus contains 147,879 AI-related sentences, including 67,080 actionable sentences, 17,900 speculative sentences, and 62,899 irrelevant sentences.

Appendix Table A1 reports the measurement audit underlying the disclosure-classification layer. The audit includes 551 adjudicated labels, a leakage-safe training and calibration pool of 431 observations, and a 120-observation adjudicated held-out benchmark. In the human–human IRR rerun, Cohen’s kappa is 0.850 after adjudication of 12 disagreements. On the held-out benchmark, the selected hybrid policy attains 85.0% accuracy, 83.6% macro-F1, 91.7% binary relevance, and 86.5% actionable-versus-speculative performance.

3.3 Firm-year disclosure measures

Write $A_{i,t}$, $S_{i,t}$, and $I_{i,t}$ for the number of actionable, speculative, and irrelevant AI sentences in firm i ’s filing in year t . The total number of AI-related sentences is then

$$N_{i,t} = A_{i,t} + S_{i,t} + I_{i,t}.$$

The broad disclosure-intensity measure is

$$\text{AIFocus}_{i,t} = \log(1 + N_{i,t}).$$

We also define disclosure-type indicators. $\text{ActionableDisclosure}_{i,t}$ equals one if $A_{i,t} > 0$ and zero otherwise. $\text{SpeculativeOnly}_{i,t}$ equals one if $S_{i,t} > 0$ and $A_{i,t} = 0$. These two indicators separate firm-years with any concrete AI implementation language from firm-years that mention AI only in speculative terms.

For descriptive work and companion specifications, we also retain the sentence counts and their log transforms. To summarize credibility tilt, we define

$$\text{ActionableShare}_{i,t} = \frac{A_{i,t}}{A_{i,t} + S_{i,t}}, \quad \text{SpecShare}_{i,t} = \frac{S_{i,t}}{A_{i,t} + S_{i,t}},$$

for firm-years with $A_{i,t} + S_{i,t} > 0$, and zero otherwise. We also define a single credibility score that rises with concrete content and falls with speculative content,

$$\text{CredAI}_{i,t} = z(A_{i,t}) - z(S_{i,t}),$$

where $z(\cdot)$ standardizes a variable to mean zero and unit standard deviation across the sample. A higher $\text{CredAI}_{i,t}$ means the firm uses more actionable and less speculative AI language than the typical firm-year.

The main metric used in the interaction specifications is the actionable-to-speculative (AS) ratio

$$AS_{i,t} = \log \left(1 + \frac{A_{i,t}}{1 + S_{i,t}} \right).$$

This stabilized ratio rises as AI language becomes more actionable and falls as speculative language becomes more prominent, while remaining defined when $S_{i,t} = 0$. In our empirical analysis, the AS ratio is the primary disclosure-credibility regressor, while $\text{SpecShare}_{i,t}$ and $\text{CredAI}_{i,t}$ are secondary companion measures.

The sample retains many zero-disclosure years by construction. For those firm-years, the intensity and composition measures equal zero. This is a feature of the design, not a data error: keeping these zero years is what makes the timing comparisons interpretable.

3.4 Outcome families and organizational response settings

The empirical design evaluates the disclosure credibility construct across innovation, market, real outcome, scrutiny, incentive, and financing window evidence. Patent outcomes validate the construct against later AI-related realization. Market and capital market outcomes study filing-date reactions, post-filing returns, valuation, and financing. Organizational response tests examine settings in which disclosure credibility may change.

The innovation outcomes are built from the matched AI patent series described above. The main patent outcome is $\log(1 + \text{AIPatents}_{i,t+h})$, where $h \in \{-2, -1, 0, +1, +2\}$ indexes calendar-time horizons relative to the disclosure year. These outcomes validate the construct against later observable technological output and separate backward-looking, contemporaneous, and forward-looking timing patterns.

The market-based outcomes are constructed from CRSP return data linked to annual 10-K filing dates. For short-window market reaction tests, we use filing-date cumulative abnormal returns computed from daily stock returns net of the value-weighted market return. The baseline filing-date window is $[-1, +1]$ trading days around the filing date, and the event-study diagnostics also plot returns over the daily window from trading day -2 to trading day $+2$. For post-filing abnormal-return regressions, we construct buy-and-hold abnormal returns beginning on trading day $+2$ so that the immediate filing window is excluded. These post-filing windows are evaluated over horizons of 1, 3, 6, and 12 months.

To complement the event-level return tests, we also form calendar-time long-short portfolios using the latest available filing signal for each stock-month. In these portfolios, the long leg holds non-mismatch AI filers and the short leg holds mismatch AI filers. We evaluate both equal-weight

and value-weight implementations over alternative holding windows. Monthly portfolio spreads are benchmarked using CAPM, Fama–French three-factor, Fama–French five-factor, and five-factor-plus-momentum models, with the main text emphasizing the richest benchmark. The portfolio tests show whether post-filing return differentials are concentrated in smaller firms and whether they survive standard factor adjustment.

The valuation and financing tests use annual, cross-sectional outcomes rather than event-time returns. The outcomes are constructed from linked market features and accounting data. The valuation side includes log market-capitalization-to-assets and a log Tobin’s- Q proxy. The financing side includes the next-year change in shares outstanding and an indicator for substantial next-year share growth. Because the main-text valuation and financing tests focus on non-big firms, broader variants are reported as supporting material.

3.5 PatentMismatch construction and determinants

The central low-credibility disclosure construct is PatentMismatch. The indicator is defined at the firm-year level and can equal one only in AI-talking firm-years. PatentMismatch identifies AI-talking firm-years whose disclosure composition is weak relative to the firm’s contemporaneous AI implementation record. The construct therefore combines a language-side signal with a same-year technology-side signal rather than relying on either one alone.

Formally, $\text{PatentMismatch}_{i,t} = 1$ when three conditions hold: firm-year (i,t) is an AI-talking year; $AS_{i,t}$ falls in the bottom quartile of the year- t AI-talking distribution or $\text{SpecShare}_{i,t}$ lies in the top quartile of that distribution; and contemporaneous $\log(1 + \text{AIPatents}_{i,t})$ is below the same-industry-year mean of $\log(1 + \text{AIPatents})$. Otherwise, $\text{PatentMismatch}_{i,t} = 0$. The disclosure-side gate uses an OR rule across the AS ratio and speculative share; the patent-weakness condition must also hold. Outside AI-talking firm-years, the indicator is coded as zero.

AI-washing refers to the broader disclosure phenomenon in which AI-related narratives appear stronger than contemporaneous or subsequent implementation. PatentMismatch is the main empirical proxy for that phenomenon, constructed from filing language and contemporaneous patent weakness. Low-credibility disclosure refers to the language-side component of this proxy.

We use two related variants in robustness and mechanism tables. LowCredibility is the language-only component of the mismatch rule: it flags AI-talking firm-years in which the AS ratio falls in the bottom quartile of the year-specific AI-talking distribution or SpecShare lies in the top quartile of that distribution, without imposing the patent-weakness condition. ApplicationMismatch applies the same low-credibility disclosure gate but replaces the patent-grant weakness condition with weakness in contemporaneous AI patent applications relative to the industry-year benchmark. These variants decompose the canonical construct by isolating the disclosure-only and application-based components.

The timing is important for interpretation. PatentMismatch uses only year- t information: the composition of disclosure in the filing year and contemporaneous patent weakness measured relative to the industry-year benchmark. It does not use future patent outcomes in its construction. Using only filing-year information avoids mechanical circularity when the construct is later related to post-filing returns or to future AI patenting. Because the patent component is based on grant timing rather than application timing, PatentMismatch should be interpreted as a filing-year disclosure-credibility construct rather than as a signal that is perfectly observable to investors at the exact filing date.

PatentMismatch enters the empirical blocks in three ways. In the annual innovation-validation regressions, it enters either directly or interacted with the AS ratio to test whether lower-credibility AI disclosure predicts weaker subsequent AI patent realization. In the filing-date and post-filing

market tests, it serves as the main low-credibility disclosure indicator used to distinguish firms whose AI language appears least aligned with contemporaneous implementation. In the annual valuation and financing tests, it is used to ask whether these same low-credibility firm-years are associated with different market valuation or financing outcomes.

A set of appendix tests asks where mismatch concentrates across firms. In those specifications, the dependent variable is either an indicator for whether a firm records at least one PatentMismatch incident during 2016–2025 or the share of its AI-talking years flagged as mismatch over the same period. Baseline firm characteristics are measured in 2016, and these tests describe where mismatch is concentrated rather than establish a causal mechanism.

3.6 Empirical specifications

The core specifications cover innovation validation, filing-date returns, post-filing returns, valuation and financing outcomes, and organizational-response tests. All specifications use the same underlying disclosure layer, but the unit of observation and timing structure differ across blocks.

First, to validate the disclosure measures against later technological output, we estimate annual panel regressions of the form

$$\log(1 + \text{AIPatents}_{i,t+h}) = \alpha_h + \beta_h \text{Narrative}_{i,t} + X'_{i,t} \theta_h + \gamma_i + \gamma_t + \varepsilon_{i,t+h},$$

where $h \in \{-2, -1, 0, +1, +2\}$ and $\text{Narrative}_{i,t}$ is alternatively $\text{AIFocus}_{i,t}$, $\text{ActionableDisclosure}_{i,t}$, $\text{SpeculativeOnly}_{i,t}$, or the AS ratio interacted with $\text{PatentMismatch}_{i,t}$. The baseline control set includes log assets, leverage, cash/assets, R&D/assets, CAPX/assets, ROA, sales growth, and employees. The main specifications include firm and year fixed effects, with companion columns using industry $\times$ year fixed effects and standard firm-level clustering. These regressions are predictive timing tests rather than causal designs. They show whether different forms of AI disclosure relate differently to prior, contemporaneous, and later AI patenting.

Second, to study immediate market reaction, we estimate filing-level event regressions of the form

$$CAR_{i,f,[a,b]} = \alpha + \beta_1 \text{PatentMismatch}_{i,t} + \beta_2 AS_{i,t} + \beta_3 \text{AIFocus}_{i,t} + X'_{i,t-1} \theta + \delta_j + \lambda_\tau + \varepsilon_{i,f},$$

where $CAR_{i,f,[a,b]}$ is the market-adjusted cumulative abnormal return for firm i around filing event f over event window $[a, b]$, δ_j denotes industry fixed effects, and λ_τ denotes filing-year fixed effects. The baseline filing window is $[-1, +1]$. The event sample is restricted to AI-talking 10-K and 10-K/A filings with complete event-window returns and successful merges back to the annual disclosure panel. Standard errors are clustered at the firm level. These tests ask whether low-credibility AI disclosure is associated with different abnormal-return patterns at the filing date, conditional on lagged firm characteristics and common industry and filing-year effects.

Third, we estimate post-filing abnormal-return regressions using buy-and-hold abnormal returns beginning on trading day $+2$:

$$BHAR_{i,f,[+2,H]} = \alpha + \beta_1 \text{PatentMismatch}_{i,t} + \beta_2 AS_{i,t} + \beta_3 \text{AIFocus}_{i,t} + X'_{i,t-1} \theta + \delta_j + \lambda_\tau + \varepsilon_{i,f,H},$$

where H indexes the post-filing horizon. We evaluate horizons corresponding to approximately 1, 3, 6, and 12 months. We complement these regressions with calendar-time long-short portfolios that buy non-mismatch AI filers and short mismatch filers using the most recent filing signal for each stock-month. Equal-weight and value-weight portfolios are evaluated separately, and the portfolio spread is benchmarked under CAPM, Fama–French 3-factor, Fama–French 5-factor, and 5-factor-plus-momentum models. These tests assess whether return differentials emerge after filing, survive factor adjustment, and concentrate in smaller firms.

Fourth, to examine annual capital-market consequences, we estimate firm-year regressions of the form

$$Y_{i,t+h} = \alpha + \beta_1 \text{PatentMismatch}_{i,t} + \beta_2 \text{AS}_{i,t} + \beta_3 \text{AIFocus}_{i,t} + X'_{i,t} \theta + \gamma_i + \gamma_t + \varepsilon_{i,t+h},$$

where $Y_{i,t+h}$ is alternatively a valuation or financing outcome. The valuation outcomes include log market-capitalization-to-assets and a log Tobin's- Q proxy, and the financing outcomes include next-year share growth and an indicator for substantial next-year issuance. These specifications use firm- and year fixed effects with firm-clustered standard errors. The main-text versions report the non-big subsample, with broader variants retained as supporting evidence.

The same disclosure layer also enters organizational-response specifications reported after the main validation tests. These specifications are descriptive event-window or panel designs rather than a primary identification strategy. The comment-letter tests compare disclosure-composition changes around each continuing AI discloser's first SEC comment-letter year. The capital-raising tests align disclosure dynamics around each firm's first large equity-issuance window, defined using next-year CRSP shares outstanding growth above 5%. The executive-incentive tests link AI-talking firm-years to lagged CEO characteristics from ExecuComp. The SEC enforcement-salience tests use the March 18, 2024 AI-washing enforcement actions (SEC, 2024) as a public salience date and compare firms already exposed to AI-washing risk in 2022 with other AI-talking firms. These specifications characterize settings in which low-credibility AI disclosure becomes more or less prevalent; they are not designed to isolate a single causal mechanism.

The disclosure variables keep the same interpretation across blocks. $\text{AIFocus}_{i,t}$ captures the broad intensity of the AI narrative, the AS ratio captures the composition of the disclosure, and $\text{PatentMismatch}_{i,t}$ identifies firm-years in which disclosure composition and contemporaneous implementation appear misaligned. The specifications are interpreted as linked evidence on disclosure credibility rather than as a single causal design.

3.7 Supporting post-ChatGPT diagnostic

A supporting diagnostic in Appendix Figure C1 asks whether credibility patterns intensify after the public release of ChatGPT. We treat this exercise as exploratory. The release of ChatGPT sharply increased the salience of AI-related language and may have lowered the cost of inserting AI-related narrative into annual filings, but it also coincides with broader changes in implementation, attention, and experimentation. The diagnostic therefore serves as a sensitivity check on the post-2022 interpretation rather than as a primary identification strategy.

The baseline implementation defines $\text{PostChatGPT}_t = 1$ for the years 2023 and later and constructs treatment status from the pre-period 2018–2022. The main treatment identifies low-capability firms: firms with zero pre-period AI patents and either a low pre-period AS ratio or no pre-period actionable AI disclosure. The intuition is that these firms have the weakest ex ante evidence of AI capability yet may be the most exposed to a fall in the cost of AI-related narrative production. We also consider two alternative treatment definitions in supporting specifications: firms with high pre-period speculative intensity relative to their industry and firms with low pre-period market capitalization.

The exploratory specifications take the form of

$$Y_{i,t} = \alpha_i + \lambda_t + \beta(\text{PostChatGPT}_t \times \text{Treated}_i) + X'_{i,t-1} \theta + \varepsilon_{i,t},$$

where $Y_{i,t}$ is alternatively AI Focus, AS ratio, PatentMismatch, filing-date abnormal returns, or post-filing abnormal returns. The annual specifications use firm- and year fixed effects where appropriate,

and the event-based outcome specifications inherit the corresponding event-study sample restrictions. These tests examine whether low-capability firms show sharper increases in low-credibility AI disclosure after late 2022.

The post-ChatGPT specifications are reported as supporting diagnostics rather than as the main identification design. The corresponding table and appendix figure report outcome-specific pretrend diagnostics, which are important for interpreting whether the post-2022 patterns are more consistent with disclosure-composition shifts than with a parallel filing-date market response.

4 Empirical Analysis and Results

4.1 The scale of low-credibility AI disclosure

Figure 1 documents the central descriptive pattern. Low-credibility AI disclosure rises sharply after 2022. The number of AI-talking firm-years flagged as PatentMismatch rises sharply after that date, and the mismatch share among AI-talking firm-years reaches 41.48% in 2024 and 42.12% in 2025. The expansion of AI language is not limited to disclosure volume. It also reflects a rise in firm-years where disclosure composition is inconsistent with contemporaneous AI patenting.

Table 1 describes the annual sample that underlies the main tests. The panel contains 50,840 firm-year observations across 5,084 firms, of which 13,777 are AI-talking firm-years, and the classified corpus contains 147,879 AI-related sentences. AI-related disclosure is highly skewed, while AI patenting remains concentrated in a much smaller subset of firms. The surge in AI-related disclosure occurs while visible implementation remains uneven.

4.2 Validation against later AI realization

Validation begins with later AI realization. Table 2 compares alternative construct definitions. The canonical grant-based mismatch measure predicts weaker subsequent AI patent grants at $t + 1$ with a coefficient of -0.0435 ($p < 0.01$), whereas a stricter mismatch variant is weaker and unstable. A weak-patent-only indicator is stronger for near-term grants by construction, but it is not a substitute for PatentMismatch because it removes the disclosure side of the construct. Canonical mismatch is associated with stronger AI application activity at $t + 2$, consistent with delayed realization or experimentation despite weak near-term patent grants. The construct-variant screen supports the canonical disclosure-plus-implementation measure and clarifies the limits of looser alternatives.

4.3 Real-outcome dynamics

Operating outcomes provide a second validation check. Table 3 links PatentMismatch to future ROA, R&D intensity, and related outcomes. In the AI-talking sample, PatentMismatch predicts weaker profitability at the two-year horizon and stronger later R&D intensity. The coefficient on ROA at $t + 2$ is -0.0672 ($p < 0.05$), while the coefficient on R&D-to-assets at $t + 2$ is 0.0178 ($p < 0.05$). LowCredibility and ApplicationMismatch show similar patterns. The estimates imply a clear pattern: low-credibility AI disclosure is followed by weaker near-term performance and higher subsequent R&D.

4.4 Scrutiny without silence

External scrutiny provides the first organizational-response test. Table 4 studies within-firm changes around each firm’s first SEC comment-letter year, restricting the main sample to active AI disclosers that mention AI in both $t - 1$ and $t + 1$. The key pattern is disclosure cleanup without silence. In

Table 1: Expanded-panel summary statistics and sample coverage*Panel A. Coverage and corpus totals*

Metric	Value	Notes
Sample firms	5,084	Unique firms in the annual sample.
Firm-year observations	50,840	Annual panel coverage for 2016–2025.
AI-talking firm-years	13,777	Firm-years with at least one classified AI sentence.
Total classified AI sentences	147,879	Total across the 2016–2025 annual sample.
Actionable AI sentences	67,080	Corpus total.
Speculative AI sentences	17,900	Corpus total.
Irrelevant AI sentences	62,899	Corpus total.
Firm-years with Compustat controls	31,291	Non-missing log assets in the annual panel.
Firm-years with matched market cap	23,747	Annual rows with merged CRSP market-cap data.
Filing-event observations	7,355	Distinct filing events across 2,753 firms.
Filing events with CAR[−1, +1]	7,342	Usable immediate-return events.
Filing events with BHAR[+2, +63]	7,098	Usable 3-month drift events.
Filing events with BHAR[+2, +252]	4,756	Usable 12-month drift events.

Panel B. Annual-panel variables

Variable	Mean	Std. Dev.	p5	p25	p50	p75	p95	N
Total AI sentences per firm-year	2.909	17.306	0.000	0.000	0.000	1.000	13.000	50,840
Actionable AI sentences per firm-year	1.319	8.293	0.000	0.000	0.000	0.000	6.000	50,840
Speculative AI sentences per firm-year	0.352	2.494	0.000	0.000	0.000	0.000	1.000	50,840
Irrelevant AI sentences per firm-year	1.237	8.209	0.000	0.000	0.000	0.000	6.000	50,840
AI Focus	0.463	0.933	0.000	0.000	0.000	0.693	2.639	50,840
Actionable share	0.129	0.305	0.000	0.000	0.000	0.000	1.000	50,840
SpecShare	0.031	0.136	0.000	0.000	0.000	0.000	0.200	50,840
CredAI	0.000	0.755	-0.298	-0.018	-0.018	-0.018	0.344	50,840
Actionable-to-speculative ratio (<i>AS</i>)	0.210	0.516	0.000	0.000	0.000	0.000	1.386	50,840
PatentMismatch	0.096	0.295	0.000	0.000	0.000	0.000	1.000	50,840
AI patents	0.160	2.969	0.000	0.000	0.000	0.000	0.000	50,840
Total patents	6.012	76.473	0.000	0.000	0.000	0.000	10.000	50,840
Log assets	6.981	2.799	2.096	5.361	7.317	8.889	10.992	31,291
Leverage	0.241	0.253	0.000	0.026	0.185	0.371	0.708	31,186
Cash/assets	0.238	0.275	0.004	0.037	0.114	0.352	0.879	31,290
R&D/assets	0.164	0.309	0.000	0.010	0.060	0.187	0.639	17,934
CAPX/assets	0.028	0.039	0.000	0.002	0.014	0.038	0.104	31,113
ROA	-0.232	0.912	-1.199	-0.147	0.010	0.049	0.150	31,228
Sales growth	1.310	31.591	-0.378	-0.025	0.072	0.212	1.016	28,010
Employees	13.495	59.888	0.007	0.182	1.400	7.600	57.000	29,404
Log MktCap/assets	6.811	1.187	4.675	6.088	6.894	7.638	8.679	23,444
Log Q proxy	6.809	1.186	4.675	6.087	6.890	7.635	8.673	23,369
$\Delta \log Q \ t + 1$	-0.062	0.541	-1.057	-0.305	-0.024	0.216	0.757	19,894
$\Delta \text{ shares } t + 1$	0.083	0.407	-0.097	-0.010	0.006	0.049	0.562	20,391
Issue > 5% $t + 1$	0.247	0.431	0.000	0.000	0.000	0.000	1.000	20,391

Panel C. Event-study and market variables

Variable	Mean	Std. Dev.	p5	p25	p50	p75	p95	N
Total AI sentences per filing	7.205	17.075	1.000	1.000	2.000	7.000	28.000	7,355
Actionable AI sentences per filing	3.875	9.640	0.000	0.000	1.000	4.000	16.000	7,355
Speculative AI sentences per filing	0.878	2.602	0.000	0.000	0.000	1.000	4.000	7,355
Actionable share per filing	0.545	0.415	0.000	0.000	0.600	1.000	1.000	7,355
Speculative share per filing	0.123	0.254	0.000	0.000	0.000	0.125	1.000	7,355
CAR[−1, +1]	0.000	0.116	-0.159	-0.037	-0.001	0.033	0.159	7,342
BHAR[+2, +21]	-0.022	0.219	-0.315	-0.109	-0.023	0.051	0.244	7,265
BHAR[+2, +63]	-0.036	0.380	-0.499	-0.204	-0.050	0.090	0.430	7,098
BHAR[+2, +126]	-0.064	0.530	-0.695	-0.304	-0.084	0.111	0.572	6,829
BHAR[+2, +252]	-0.043	1.384	-0.902	-0.452	-0.135	0.151	0.919	4,756

Note. Panel A reports sample coverage and corpus totals. Panels B and C report distributional statistics for annual-panel and filing-event variables. Sentence totals in Panel A are corpus totals; sentence variables in Panels B and C are per-firm-year or per-filing quantities.

Table 2: Construct-variant screen for future AI realization

<i>Panel A. Future AI patent-grant outcomes</i>		
Variant	Log(1 + AI patent grants) t+1	Log(1 + AI patent grants) t+2
Canonical grant mismatch	-0.0435*** (0.0134)	0.0085 (0.0081)
Strict grant mismatch	-0.0274 (0.0222)	-0.0301 (0.0187)
Application mismatch	-0.0308*** (0.0092)	0.0062 (0.0085)
Low credibility only	-0.0034 (0.0131)	0.0121 (0.0138)
Weak patent only	-0.1276*** (0.0239)	0.0020 (0.0152)
Outcome mean	0.104	0.092
Observations	10,129	10,129
<i>Panel B. Future AI application outcomes</i>		
Variant	Log(1 + AI applications) t+1	Log(1 + AI applications) t+2
Canonical grant mismatch	0.0221* (0.0114)	0.0685*** (0.0141)
Strict grant mismatch	-0.0130 (0.0149)	-0.0032 (0.0167)
Application mismatch	-0.0100 (0.0086)	0.0157* (0.0089)
Low credibility only	0.0084 (0.0134)	0.0185 (0.0154)
Weak patent only	0.0521*** (0.0182)	0.1638*** (0.0232)
Outcome mean	0.140	0.118
Observations	10,129	10,129

Note. This table compares alternative low-credibility disclosure constructs against future AI realization outcomes. Panel A uses future AI patent grants and Panel B uses future AI patent applications. All outcomes are measured at annual horizons relative to the disclosure year. Strict grant mismatch requires both a low AS ratio and a high speculative share before applying the patent-weakness condition; application mismatch replaces grant weakness with AI patent-application weakness; low credibility only uses the language gate alone; weak patent only uses the patent-weakness condition without the disclosure gate. The table is intended to show why the canonical grant-based PatentMismatch measure is retained for the main analysis. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Table 3: Real-outcome dynamics in the AI-talking sample

Outcome	Canonical grant mismatch	Application mismatch	Low credibility only	Outcome mean	Observations
ROA t+1	-0.0045 (0.0157)	0.0003 (0.0156)	-0.0083 (0.0151)	-0.224	6,865
ROA t+2	-0.0672** (0.0269)	-0.0526** (0.0267)	-0.0649*** (0.0248)	-0.223	4,476
Sales growth t+1	0.0231 (0.0502)	-0.0297 (0.0473)	-0.0065 (0.0430)	0.240	6,532
Sales growth t+2	-0.0240 (0.0550)	0.0599 (0.0553)	0.0550 (0.0460)	0.211	4,284
CAPX/assets t+1	-0.0003 (0.0009)	-0.0009 (0.0008)	-0.0006 (0.0008)	0.024	6,829
CAPX/assets t+2	0.0012 (0.0010)	0.0013 (0.0009)	0.0014 (0.0010)	0.023	4,449
R&D/assets t+1	0.0097 (0.0070)	0.0079 (0.0073)	0.0087 (0.0064)	0.149	4,514
R&D/assets t+2	0.0178** (0.0070)	0.0154** (0.0074)	0.0166*** (0.0062)	0.147	3,044

Note. This table reports forward operating and investment outcomes in the AI-talking annual sample. Coefficients are reported for the canonical grant-based PatentMismatch construct and alternative construct variants. Outcomes are measured one and two years after the disclosure year. Standard errors are shown in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

the main sample, the AS ratio rises by 0.1796 at $t + 1$ ($p = 0.007$), SpecShare falls by 0.1051 at $t + 2$ ($p = 0.025$), and PatentMismatch declines by 0.1143 at $t + 1$ ($p = 0.073$). AI Focus also rises after scrutiny rather than falling. The effect is not driven by firms dropping AI language from their filings. Continuing AI disclosers rebalance toward more credible composition after scrutiny. The AI-specific comment subset is small, so the broader sample drives the estimates.

Table 4: Disclosure cleanup after SEC comment-letter scrutiny

Outcome	t-2 vs t-1	t vs t-1	t+1 vs t-1	t+2 vs t-1	Pre mean	N firms
<i>Panel A. First SEC comment letter (active AI disclosers)</i>						
Speculative share	-0.1799*** (0.0519)	-0.0476 (0.0304)	-0.0621 (0.0407)	-0.1051** (0.0457)	0.2717	70
AS ratio	-0.1768*** (0.0644)	0.0466 (0.0599)	0.1796*** (0.0651)	0.2213** (0.1049)	0.7910	70
PatentMismatch	-0.1935*** (0.0684)	-0.0429 (0.0516)	-0.1143* (0.0629)	-0.1207* (0.0699)	0.4143	70
AI Focus	-0.5977*** (0.0920)	0.1993** (0.0794)	0.5344*** (0.0964)	0.5574*** (0.1409)	1.4720	70
<i>Panel B. First AI-related SEC comment letter (active AI disclosers)</i>						
Speculative share	-0.0097 (0.0750)	0.0241 (0.0543)	0.1360 (0.1034)	0.0697 (0.1369)	0.1471	11
AS ratio	-0.1434 (0.1171)	0.1883 (0.1213)	-0.0054 (0.1842)	0.1767 (0.2122)	0.8552	11
PatentMismatch	-0.0909 (0.2113)	0.0909 (0.0909)	0.2727* (0.1408)	0.1250 (0.2266)	0.2727	11
AI Focus	-0.4144** (0.1367)	0.6666*** (0.1303)	0.8162** (0.3076)	1.1843** (0.3764)	2.0795	11

Note. Each cell reports the mean within-firm change relative to the filing year immediately before the first comment-letter event ($t - 1$) among active AI disclosers, defined as firms that mention AI in both $t - 1$ and $t + 1$. Standard errors are based on paired differences across firms. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

4.5 Regulatory salience and disclosure cleanup

A second scrutiny test centers on the SEC’s March 18, 2024 AI-washing enforcement actions (SEC, 2024). Table 5 fixes treatment using firms that were already exposed in 2022. Pre-exposed firms become less speculative and less likely to be flagged as mismatch after 2024. The post-2024 coefficient is -0.0839 for SpecShare, 0.2907 for the AS ratio, and -0.3121 for PatentMismatch, all with $p < 0.001$. The same pattern appears under alternative exposure definitions based on LowCredibility and ApplicationMismatch. The 2023 placebo estimates are already directionally similar for the composition outcomes, limiting a sharp enforcement interpretation. The post-2024 estimates are better read as regulatory salience or acceleration in disclosure cleanup. Substantively, disclosure composition becomes more credible after the SEC AI-washing salience event without a comparable collapse in AI Focus.

4.6 Managerial incentives and financing-window opportunism

Managerial incentives show a related pattern. Table 6 links the AI-talking annual panel to lagged CEO observations from ExecuComp. In the post-ChatGPT sample, PatentMismatch is positively associated with lagged CEO equity-award share and CEO ownership, with coefficients of 0.1787 ($p = 0.041$) and 0.0170 ($p = 0.036$), respectively. In the big-firm sample, PatentMismatch is also positively associated with CEO ownership, with a coefficient of 0.1038 ($p < 0.001$). The estimates are not causal, but PatentMismatch is more prevalent when CEO incentives are more equity-oriented and managerial ownership is higher.

Financing windows provide a sharper timing test. Table 7 aligns disclosure dynamics around each firm’s first large equity-issuance event and focuses on active AI issuers that continue discussing AI in the event year and the following year. Credibility deteriorates before the financing event and partly reverses afterward. Among all active AI issuers, LowCredibility rises by 0.2334 and

Table 5: SEC AI-washing enforcement salience and disclosure cleanup

<i>Panel A. Enforcement-salience DID using mismatch exposure in 2022</i>			
Outcome	Post-2024 DID	2023 placebo	Mean / treated
Speculative share	-0.0839*** (0.0184)	-0.1224*** (0.0187)	0.160
AS ratio	0.2907*** (0.0345)	0.3314*** (0.0315)	0.939
PatentMismatch	-0.3121*** (0.0288)	-0.4305*** (0.0284)	0.261
AI Focus	-0.1044 (0.0653)	0.0118 (0.0495)	1.850
<i>Panel B. Alternative exposure definitions on core disclosure outcomes</i>			
Treatment definition	SpecShare DID	AS ratio DID	PatentMismatch DID
LowCredibility in 2022	-0.1215*** (0.0166)	0.3368*** (0.0331)	-0.2606*** (0.0259)
ApplicationMismatch in 2022	-0.1042*** (0.0185)	0.3155*** (0.0344)	-0.2634*** (0.0290)

Note. Treatment is fixed using firms that mention AI in 2022 and satisfy the exposure condition listed in the table. The post period is 2024–2025, after the SEC’s March 18, 2024 AI-washing enforcement actions (SEC, 2024). Panel A reports the main PatentMismatch-exposure DID estimates and the 2023 placebo coefficient from the event-time specification. Panel B reports post-2024 DID coefficients for alternative exposure definitions on speculative share, AS ratio, and PatentMismatch, respectively. All specifications absorb firm and year fixed effects, include lagged controls, and cluster standard errors by firm. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Table 6: Executive incentives and low-credibility AI disclosure

Panel	Outcome	Equity-award share	Ownership pct	Log CEO pay	Outcome mean	Obs.
<i>Panel A. ExecuComp-linked AI-talking sample</i>						
	PatentMismatch	0.0035 (0.0628)	0.0052 (0.0101)	-0.0021 (0.0190)	0.374	979
	LowCredibility	-0.0179 (0.0790)	-0.0008 (0.0084)	-0.0100 (0.0225)	0.543	979
	AS ratio	0.0896 (0.1189)	-0.0021 (0.0144)	0.0677* (0.0350)	0.785	979
<i>Panel B. Big-firm ExecuComp sample</i>						
	PatentMismatch	0.0322 (0.0672)	0.1038*** (0.0290)	0.0003 (0.0198)	0.375	771
	LowCredibility	0.0002 (0.0912)	0.0440 (0.0507)	-0.0108 (0.0242)	0.541	771
	AS ratio	0.0215 (0.1321)	-0.1113* (0.0674)	0.0605 (0.0377)	0.803	771
<i>Panel C. Post-ChatGPT ExecuComp sample</i>						
	PatentMismatch	0.1787** (0.0874)	0.0170** (0.0081)	0.0528* (0.0321)	0.442	407
	LowCredibility	0.2036 (0.1247)	0.0290*** (0.0065)	0.0709 (0.0462)	0.580	407
	AS ratio	-0.2022 (0.1255)	-0.0369*** (0.0069)	-0.0687 (0.0512)	0.705	407

Note. The sample links the AI-talking annual panel to lagged CEO observations from ExecuComp, using the highest-TDC1 CEO row when multiple CEO-designated rows exist in a firm-year. Predictors are one-year lagged CEO equity-award share of total compensation, CEO ownership percentage excluding options, and log total CEO pay. Specifications absorb firm and year fixed effects, include the baseline annual control set, and cluster standard errors by firm. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

PatentMismatch rises by 0.1787 into the issue year, then both decline in the following year. AI Focus, by contrast, continues to rise through and after the issue year. The non-big subsample shows the same timing shape. The timing is hard to attribute to silence or noise alone. The evidence is more consistent with AI promotion around financing events followed by partial cleanup.

Table 7: Capital-raising timing and low-credibility AI disclosure

Panel	Outcome	Event year vs. t-1	t+1 vs. event year	t+2 vs. event year	Event-year mean	Obs.
<i>Panel A. First large equity issue among active AI issuers</i>						
	LowCredibility	0.2334*** (0.0294)	-0.1063*** (0.0233)	-0.1499*** (0.0297)	0.4741	367
	ApplicationMismatch	0.1816*** (0.0276)	-0.0790*** (0.0242)	-0.1008*** (0.0288)	0.3597	367
	PatentMismatch	0.1787*** (0.0269)	-0.0926*** (0.0242)	-0.1144*** (0.0285)	0.3678	367
	Speculative share	0.1187*** (0.0158)	-0.0177 (0.0118)	-0.0212 (0.0160)	0.1813	367
	AI Focus	1.0051*** (0.0545)	0.3962*** (0.0410)	0.4644*** (0.0592)	1.6775	367
<i>Panel B. Non-big first-issue AI issuers</i>						
	LowCredibility	0.2250*** (0.0344)	-0.0956*** (0.0276)	-0.1355*** (0.0360)	0.4382	251
	ApplicationMismatch	0.1875*** (0.0335)	-0.0797*** (0.0289)	-0.1076*** (0.0348)	0.3705	251
	PatentMismatch	0.1708*** (0.0323)	-0.0797*** (0.0289)	-0.1036*** (0.0342)	0.3586	251
	Speculative share	0.1231*** (0.0189)	-0.0171 (0.0144)	-0.0305 (0.0190)	0.1832	251
	AI Focus	1.0540*** (0.0663)	0.3882*** (0.0507)	0.4174*** (0.0766)	1.6373	251

Note. Event year 0 is the disclosure year preceding a large equity-issuance window, where $\text{Issue} > 5\%$ is defined as next-year CRSP shares-outstanding growth above 5%. The event sample keeps each firm's first such issue year and focuses on active AI issuers that continue talking about AI in the event year and the following year. Positive values in the first comparison indicate deterioration or intensification into the issuance window; negative values in the second and third comparisons indicate partial unwind after the issue year. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

4.7 Mixed and benchmark-sensitive market evidence

The market-pricing evidence is more limited than the validation, scrutiny, and incentive evidence. Filing-date abnormal returns remain muted, so the data do not support a clean immediate pricing response at the annual 10-K filing date. Table 8 reports the benchmark-adjusted portfolio results. The strongest return result is the equal-weight three-month long-credible, short-mismatch portfolio, which delivers a FF5+momentum alpha of 0.976 percentage points per month ($p = 0.026$). The corresponding value-weight specification is effectively zero. The return result is narrow and concentrated outside the largest firms.

Appendix results show that the short-run return relation depends on financing state. In those specifications, mismatch BHAR is materially less negative inside financing windows than outside them. Market pricing remains mixed and sensitive to benchmarks. The evidence is consistent with limited and state-dependent pricing of disclosure credibility, but it does not support a broad anomaly interpretation.

Table 8: Factor-adjusted calendar-time long-short alpha after AI filing signals

Horizon	Mean monthly	Annualized simple	CAPM alpha [p]	FF3 alpha [p]	FF5 alpha [p]	FF5+Mom alpha [p]	Months
<i>Panel A. Equal-weight long credible / short mismatch portfolios</i>							
1m	-0.508	-6.102	-0.218 [0.807]	0.191 [0.829]	0.183 [0.831]	0.345 [0.675]	98
3m	0.873	10.475	0.934 [0.036]	1.043 [0.023]	0.965 [0.032]	0.976 [0.026]	105
6m	-0.033	-0.393	-0.015 [0.951]	-0.020 [0.941]	-0.003 [0.990]	-0.072 [0.803]	106
12m	-0.195	-2.342	-0.198 [0.246]	-0.170 [0.326]	-0.173 [0.333]	-0.176 [0.322]	106
<i>Panel B. Value-weight long credible / short mismatch portfolios</i>							
1m	0.062	0.741	0.061 [0.943]	0.122 [0.887]	0.052 [0.950]	0.129 [0.871]	98
3m	-0.138	-1.658	-0.007 [0.986]	0.024 [0.954]	0.053 [0.897]	0.001 [0.997]	105
6m	0.045	0.542	-0.137 [0.666]	-0.089 [0.774]	-0.047 [0.885]	-0.032 [0.919]	106
12m	-0.185	-2.218	-0.364 [0.070]	-0.266 [0.167]	-0.198 [0.290]	-0.158 [0.410]	106

Note. This table reports calendar-time long–short portfolio returns using the latest available annual filing signal for each stock-month. The long leg holds non-mismatch AI filers and the short leg holds mismatch AI filers. Panel A reports equal-weight portfolios and Panel B reports value-weight portfolios. Alpha columns report monthly intercepts from CAPM, Fama–French three-factor, Fama–French five-factor, and five-factor-plus-momentum regressions; p-values are shown in brackets.

The appendix reports fuller market checks, issue-window refinements, additional governance and intermediary tests, and measurement and sample-construction diagnostics.

5 Conclusion

Annual 10-K filings help separate more credible AI disclosure from narrative positioning. The classification of AI-related disclosure in 10-K filings is scalable and based on audited reporting content. The main construct, PatentMismatch, combines low-credibility disclosure composition with contemporaneous patent weakness. In the 2016–2025 sample, low-credibility AI disclosure rises sharply after 2022; for a substantial subset of firms, AI language increases more rapidly than observable patenting activity.

Across alternative definitions, the filing-based mismatch measure predicts weaker subsequent AI patent grants. This specification is the most robust version of the construct across tests. In the AI-talking sample, mismatch firm-years subsequently exhibit lower profitability and higher R&D intensity. The patterns are more consistent with delayed realization and subsequent adjustment in real activity than with purely non-informative disclosure.

Low-credibility disclosure also varies with scrutiny and incentives. Among continuing AI disclosers, disclosure composition becomes more credible following SEC comment-letter scrutiny, while AI focus remains largely unchanged. The SEC’s March 18, 2024 AI-washing enforcement actions also coincide with a broader salience result (SEC, 2024). The March 2024 evidence is better read as regulatory salience or acceleration than as a sharply identified enforcement effect. Low-credibility AI disclosure is also more common when CEO equity incentives are stronger. Around large equity-issuance windows, disclosure credibility deteriorates prior to issuance and partly reverts afterward. Low-credibility AI disclosure appears shaped by both monitoring and opportunistic incentives.

The market-pricing results are more limited. Filing-date returns do not cleanly separate mismatch from non-mismatch firms, and post-filing return evidence remains benchmark-sensitive. The strongest return result is a short-horizon equal-weight alpha, with no comparable value-weight evidence. The issue-window evidence suggests a state-dependent rather than uniform short-run return pattern. Market tests therefore support a narrower interpretation than a broad AI-washing anomaly. The stronger evidence is that the filing-based credibility measure predicts later technological real-

ization and organizational behavior, even when market pricing remains incomplete and uneven.

AI-related narratives appear to diffuse faster than real AI implementation, and the gap is widest exactly where it is hardest to check: small firms, financing windows, and the period right after ChatGPT lowered the cost of AI language. What we cannot yet say is whether investors are fooled in a way that moves prices for long. Filing-date returns do not separate the firms, the post-filing evidence is confined to small stocks, and the value-weighted spreads are flat. Whether AI washing is mostly a credibility problem that resolves through later realization, or also a pricing problem that a sharper design would uncover, is the open question we leave for further work.

Appendix A Measurement audit and sample construction diagnostics

This appendix reports the measurement audit and sample-construction diagnostics referenced in Sections 3.1 and 3.2. Table A1 reports the sentence-labeling and classifier audit, Table A2 maps sample attrition across empirical blocks, Tables A3 and A4 report high-confidence classifier robustness checks, and Tables A5–A7 define the main paper-facing and construction variables.

Table A1: Measurement audit of the disclosure-classification layer

<i>Panel A. Human labeling and validation</i>				
Item	N	Primary metric	Supporting metric	Notes
Adjudicated sentence base	551	Human-reviewed labels	Full labeled corpus	Used to build and refine the sentence classifier.
Training and calibration pool	431	Human-reviewed labels	Excludes held-out items	Training and calibration set after removing the held-out benchmark sentences.
Held-out benchmark	120	Final held-out labels	Evaluation benchmark	Balanced benchmark used to evaluate the classifier.
Human–human reliability	120	Cohen’s kappa = 0.850	12 of 12 disagreements adjudicated	By-class kappa: actionable 0.852; speculative 0.790; irrelevant 0.907.
<i>Panel B. Classifier evaluation on the held-out benchmark</i>				
Item	N	Primary metric	Supporting metric	Notes
Local model only	120	Accuracy = 78.3%	Macro-F1 = 77.1%	Binary relevance 90.0%; actionable-vs-speculative 78.4%.
Selected hybrid model	120	Accuracy = 85.0%	Macro-F1 = 83.6%	Binary relevance 91.7%; actionable-vs-speculative 86.5%.

Note. This table reports the measurement audit for the sentence-classification step. Panel A summarizes the labeled sentence base, the training and calibration pool, the held-out benchmark, and the human–human reliability check. Panel B reports accuracy and related metrics on the held-out benchmark for the local model and for the selected hybrid model used in the paper.

Table A2: Attrition map for the expanded 2016–2025 AI-washing sample

<i>Panel A. Annual sample (firms that ever mention AI)</i>			
Empirical block	N	Share of parent	Comments / main reason for shrinkage
Full annual sample	50,840	100.0% of annual panel	5,084 unique firms across 2016–2025.
AI-talking firm-years	13,777	27.1% of annual panel	Disclosure-composition and PatentMismatch are economically meaningful here.
Rows with core annual controls	31,006	61.0% of annual panel	Requires ln assets, leverage, cash, capx/assets, and ROA.
Rows with R&D/assets	17,934	35.3% of annual panel	Most restrictive accounting variable in the annual panel.
Rows with CRSP market-cap linkage	23,747	46.7% of annual panel	Needed for valuation and financing tests.
Rows with next-year share growth	20,391	40.1% of annual panel	Requires current and next-year shares outstanding.
Non-big rows inside market-linked sample	11,876	50.0% of market-linked rows	Matched-sample yearly median market-cap split used in the non-big refinement.
<i>Panel B. Filing-event sample</i>			
Empirical block	N	Share of parent	Comments / main reason for shrinkage
Filing-event estimation sample	7,355	100.0% of event sample	Annual 10-K filing events with matched daily return histories.
CAR[-1,+2] complete	7,344	99.9% of event sample	Immediate filing-window CAR sample after daily-return matching.
BHAR[+2,+21] complete	7,274	98.9% of event sample	Requires one trading month of post-filing returns.
BHAR[+2,+63] complete	7,108	96.6% of event sample	Three-month horizon drops later-sample filings mechanically.
BHAR[+2,+126] complete	6,845	93.1% of event sample	Six-month horizon drops more late-sample filings.
BHAR[+2,+252] complete	4,774	64.9% of event sample	One-year horizon is the main event-study attrition screen.
<i>Panel C. Baseline cross-section for legacy determinants</i>			
Empirical block	N	Share of parent	Comments / main reason for shrinkage
2016 baseline cross-section	5,084	100.0% of baseline cross-section	One row per sample firm in 2016.
2016 rows with core controls	2,652	52.2% of baseline cross-section	Baseline multivariate sample with basic accounting controls present.
2016 rows with R&D/assets	1,466	28.8% of baseline cross-section	Primary pinch point for the fuller determinants appendix tables.
2016 rows with employment	2,479	48.8% of baseline cross-section	Second major source of cross-sectional sample reduction.

Note. This table maps sample changes across the main empirical blocks. The differences in sample size are mechanical consequences of the annual timing design, complete-case control requirements, CRSP availability in event windows, and the shift from firm-year panels to the 2016 firm-level cross-section in the determinants appendix.

Table A3: Classifier-risk robustness: high-confidence composition tests

Variable	Dependent variable: $\log(1 + \text{AI patents})$				
	t-2 (1)	t-1 (2)	t (3)	t+1 (4)	t+2 (5)
<i>Panel A. Actionable disclosure</i>					
Actionable disclosure (dummy)	0.047 (0.029)	0.049* (0.029)	0.030 (0.029)	0.035 (0.039)	0.072 (0.047)
<i>Panel B. Speculative-only disclosure</i>					
Speculative-only disclosure (dummy)	-0.014 (0.049)	-0.010 (0.046)	-0.006 (0.044)	-0.076 (0.080)	-0.129 (0.082)
Controls	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y
Adj. R^2	-0.953	-0.954	-0.928	-0.922	-0.916
Observations	2,810	2,810	2,810	2,810	2,810

Note. This table reports composition-based robustness checks using only sentences the classifier labels with high confidence. The dependent variable is $\log(1 + \text{AI patents})$, and the five columns report the relation at event times $t - 2$, $t - 1$, t , $t + 1$, and $t + 2$ relative to the disclosure year, so the columns trace how the disclosure indicator lines up with past, current, and future AI patenting. The purpose is to check whether the main patterns hold when the disclosure measures are restricted to the most confidently classified sentences. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Table A4: Classifier-risk robustness: high-confidence mismatch tests

Variable	Dependent variable: $\log(1 + \text{AI patents at } t+1)$			
	Firm + year FE (1)	Industry×year FE (2)	Firm + year FE, non-fin. (3)	Firm + year FE, non-fin./non-util. (4)
AS ratio	0.085* (0.049)	0.267*** (0.046)	0.087* (0.052)	0.087* (0.052)
AS ratio × PatentMismatch	-0.065 (0.040)	-0.408*** (0.054)	-0.068 (0.042)	-0.068 (0.042)
Controls	Y	Y	Y	Y
Firm FE	Y	N	Y	Y
Industry×Year FE	N	Y	N	N
Year FE	Y	N	Y	Y
Non-fin.	N	N	Y	Y
No util.	N	N	N	Y
Adj. R^2	-0.909	0.081	-0.885	-0.883
Observations	2,810	2,810	2,632	2,630

Note. This table reports mismatch-based robustness checks using only sentences the classifier labels with high confidence. The dependent variable is $\log(1 + \text{AI patents})$ at $t + 1$ throughout. The four columns differ only in the fixed-effect and sample choices: column (1) uses firm and year fixed effects on the full sample; column (2) uses industry×year fixed effects; column (3) repeats column (1) excluding financial firms; and column (4) further excludes utilities. Comparing the columns shows whether the AS-ratio and mismatch patterns survive alternative absorption and sample screens. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Table A5: Variable definitions: disclosure and credibility measures

Group	Variable	Definition
Disclosure counts	Total AI sentences	Count of classified AI-related sentences in the firm's annual filing for year t .
Disclosure counts	Actionable AI sentences	Count of AI sentences labeled actionable in year t .
Disclosure counts	Speculative AI sentences	Count of AI sentences labeled speculative in year t .
Disclosure counts	Irrelevant AI sentences	Count of AI sentences labeled irrelevant in year t .
Disclosure composition	AI Focus	Broad AI-disclosure intensity, measured as $\log(1 + \text{total AI sentences})$.
Disclosure composition	Actionable share	Actionable AI sentences divided by the sum of actionable and speculative AI sentences in the firm-year.
Disclosure composition	Speculative share	Speculative AI sentences divided by the sum of actionable and speculative AI sentences in the firm-year.
Disclosure composition	CredAI	Credibility score that rises with actionable content and falls with speculative content, computed as the standardized actionable count minus the standardized speculative count.
Disclosure composition	AS ratio	$\log(1 + A/(1 + S))$, where A and S are actionable and speculative counts; higher values mean more actionable relative to speculative disclosure.
Credibility construct	PatentMismatch	Indicator equal to one for AI-talking firm-years that are both low-credibility in disclosure and weak in contemporaneous AI patent grants relative to the industry-year mean.
Credibility construct	LowCredibility	Indicator equal to one for AI-talking firm-years with low disclosure credibility, defined as bottom yearly quartile of the AS ratio or top yearly quartile of speculative share.
Credibility construct	ApplicationMismatch	Indicator equal to one for AI-talking firm-years that satisfy the low-credibility disclosure condition and are weak in contemporaneous AI patent applications relative to the industry-year benchmark.
Credibility construct	Strict patent mismatch	Indicator equal to one for AI-talking firm-years that satisfy both language-side low-credibility conditions and are weak in contemporaneous AI patent grants relative to the industry-year benchmark.
Credibility construct	Weak patent only	Indicator equal to one for firm-years with contemporaneous AI patent grants below the industry-year benchmark, before imposing the disclosure-credibility condition.

Note. This table defines the disclosure-count, disclosure-composition, and credibility-construct variables used in the analysis. All variables are measured on the annual sample of firms that mention AI at least once during 2016–2025.

Table A6: Variable definitions: patents and accounting controls

Group	Variable	Definition
Patents	AI patent grants	Count of AI-related granted patents linked to the firm-year.
Patents	Total patents	Count of all patents linked to the firm-year.
Patents	Future AI patent grants $t+1$	$\log(1 + \text{AI patent grants})$ measured one year after the disclosure year.
Patents	Future AI patent grants $t+2$	$\log(1 + \text{AI patent grants})$ measured two years after the disclosure year.
Controls	Log assets	Natural log of total assets.
Controls	Leverage	Debt scaled by assets.
Controls	Cash/assets	Cash holdings scaled by assets.
Controls	R&D/assets	Research and development expense scaled by assets.
Controls	CAPX/assets	Capital expenditures scaled by assets.
Controls	ROA	Return on assets.
Controls	Sales growth	Year-over-year sales growth rate.
Controls	Employees	Employee count from Compustat.

Note. This table defines the patent variables and accounting controls used across the annual-panel specifications. All variables are measured on the annual sample of firms that mention AI at least once during 2016–2025.

Table A7: Variable definitions: event, valuation, financing, and treatment variables

Group	Variable	Definition
Event-study outcomes	CAR[-1,+1]	Cumulative abnormal return over the one-day-before to one-day-after filing window.
Event-study outcomes	BHAR[+2,+21]	Buy-and-hold abnormal return from trading day +2 through +21 after the filing date.
Event-study outcomes	BHAR[+2,+63]	Buy-and-hold abnormal return from trading day +2 through +63 after the filing date.
Event-study outcomes	BHAR[+2,+126]	Buy-and-hold abnormal return from trading day +2 through +126 after the filing date.
Event-study outcomes	BHAR[+2,+252]	Buy-and-hold abnormal return from trading day +2 through +252 after the filing date.
Valuation and financing	Log MktCap/assets	$\log(1 + \text{market capitalization/assets})$, using CRSP year-end market cap and Compustat assets.
Valuation and financing	Log Q proxy	$\log(1 + \text{market cap/assets} + \text{leverage})$, used when full book-equity fields are unavailable.
Valuation and financing	Delta log Q $t+1$	One-year-ahead change in the log Q proxy.
Valuation and financing	Delta shares $t+1$	Next-year growth in CRSP shares outstanding.
Valuation and financing	Issue > 5% $t+1$	Indicator equal to one when next-year shares outstanding grow by more than 5%.
Treatment and screens	PostChatGPT	Indicator for fiscal years 2023 onward in the ChatGPT-shock designs.
Treatment and screens	Non-big screen	Indicator equal to one when a firm’s market capitalization is at or below the matched-sample yearly median.

Note. This table defines the filing-event outcomes, valuation and financing variables, and treatment or sample screens used in the market, financing, and post-ChatGPT diagnostics. Event-study outcomes are measured on the filing-event sample; valuation and financing variables on the annual sample merged with CRSP.

Appendix B Supplementary validation and patent-timing results

This appendix reports supplementary validation and patent-timing results that support the main disclosure-credibility design. Table B1 reports broad AI-disclosure intensity and patent timing, Tables B2 and B3 separate actionable and speculative disclosure timing, and Table B4 reports the longer-horizon AS-ratio interaction design.

Negative adjusted R^2 values can arise in these sparse fixed-effect specifications because the adjusted statistic penalizes the fixed-effect and control structure relative to the remaining within-sample variation. The tables are reported for coefficient timing and sign patterns rather than model-fit interpretation.

Table B1: AI disclosure intensity and AI patent timing

Variable	Dependent variable: $\log(1 + \text{AI patents})$				
	$t-2$	$t-1$	t	$t+1$	$t+2$
	(1)	(2)	(3)	(4)	(5)
AI Focus	0.044*** (0.007)	0.054*** (0.008)	0.061*** (0.008)	0.045*** (0.007)	0.012** (0.005)
Controls	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y
Adj. R^2	-0.150	-0.142	-0.133	-0.152	-0.171
Observations	14,968	14,968	14,968	14,968	14,968

Note. This table presents annual panel regressions relating broad AI disclosure intensity (AI Focus) to AI patenting. The dependent variable is $\log(1 + \text{AI patents})$, and the five columns report the relation at event times $t - 2$ through $t + 2$ relative to the disclosure year. It provides the broad-intensity validation that complements the composition-based evidence in the main text. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Table B2: Actionable disclosure and AI patent timing

Variable	Dependent variable: Actionable disclosure			
	Firm + year FE	Industry + year FE	Firm + year FE, non-fin.	Firm + year FE, non-fin./non-util.
	(1)	(2)	(3)	(4)
$\log(1 + \text{AI patents})$ at $t-2$	0.018 (0.018)	0.015 (0.020)	0.013 (0.017)	0.013 (0.017)
$\log(1 + \text{AI patents})$ at $t-1$	0.021 (0.016)	0.023 (0.016)	0.024 (0.016)	0.024 (0.016)
$\log(1 + \text{AI patents})$ at t	0.033** (0.017)	0.040** (0.016)	0.030* (0.017)	0.030* (0.017)
$\log(1 + \text{AI patents})$ at $t+1$	0.018 (0.012)	0.023** (0.011)	0.019* (0.011)	0.019* (0.011)
$\log(1 + \text{AI patents})$ at $t+2$	0.050*** (0.011)	0.057*** (0.014)	0.048*** (0.012)	0.048*** (0.012)
Controls	Y	Y	Y	Y
Firm FE	Y	N	Y	Y
Industry FE	N	Y	N	N
Year FE	Y	Y	Y	Y
Non-fin.	N	N	Y	Y
No util.	N	N	N	Y
Adj. R^2	-0.160	0.016	-0.163	-0.163
Observations	14,968	14,968	13,759	13,737

Note. This table relates actionable disclosure to AI patenting across event times $t - 2$ through $t + 2$ (the table rows). The four columns differ in fixed effects and sample: column (1) uses firm and year fixed effects; column (2) uses industry and year fixed effects; column (3) repeats column (1) excluding financial firms; and column (4) further excludes utilities. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Table B3: Speculative-only disclosure and AI patent timing

Variable	Dependent variable: Speculative-only disclosure			
	Firm + year FE (1)	Industry + year FE (2)	Firm + year FE, non-fin. (3)	Firm + year FE, non-fin./non-util. (4)
$\log(1 + \text{AI patents})$ at $t-2$	-0.031*** (0.011)	-0.022** (0.009)	-0.030*** (0.011)	-0.030*** (0.011)
$\log(1 + \text{AI patents})$ at $t-1$	-0.012 (0.010)	-0.005 (0.010)	-0.012 (0.010)	-0.012 (0.010)
$\log(1 + \text{AI patents})$ at t	0.000 (0.010)	0.002 (0.010)	0.001 (0.010)	0.001 (0.010)
$\log(1 + \text{AI patents})$ at $t+1$	0.004 (0.007)	0.005 (0.007)	0.002 (0.007)	0.002 (0.007)
$\log(1 + \text{AI patents})$ at $t+2$	0.001 (0.007)	0.012* (0.007)	0.002 (0.007)	0.002 (0.007)
Controls	Y	Y	Y	Y
Firm FE	Y	N	Y	Y
Industry FE	N	Y	N	N
Year FE	Y	Y	Y	Y
Non-fin.	N	N	Y	Y
No util.	N	N	N	Y
Adj. R^2	-0.170	-0.002	-0.172	-0.172
Observations	14,968	14,968	13,759	13,737

Note. This table relates speculative-only disclosure to AI patenting across event times $t - 2$ through $t + 2$ (the table rows). The four columns differ in fixed effects and sample: column (1) uses firm and year fixed effects; column (2) uses industry and year fixed effects; column (3) repeats column (1) excluding financial firms; and column (4) further excludes utilities. Standard errors are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Table B4: AS ratio, PatentMismatch, and longer-horizon AI patenting

Variable	Dependent variable: $\log(1 + \text{AI patents at } t+2)$			
	Firm + year FE (1)	Industry×year FE (2)	Firm + year FE, non-fin. (3)	Firm + year FE, non-fin./non-util. (4)
AS ratio	0.021** (0.009)	0.071*** (0.017)	0.022** (0.009)	0.022** (0.009)
AS ratio × PatentMismatch	-0.008 (0.017)	-0.082*** (0.019)	-0.008 (0.018)	-0.008 (0.018)
Controls	Y	Y	Y	Y
Firm FE	Y	N	Y	Y
Industry×Year FE	N	Y	N	N
Year FE	Y	N	Y	Y
Non-fin.	N	N	Y	Y
No util.	N	N	N	Y
Adj. R^2	-0.171	0.040	-0.173	-0.173
Observations	14,968	14,968	13,759	13,737

Note. This companion table uses $\log(1 + \text{AI patents})$ at $t + 2$ as the outcome. The four columns differ in fixed effects and sample: column (1) uses firm and year fixed effects; column (2) uses industry×year fixed effects; column (3) repeats column (1) excluding financial firms; and column (4) further excludes utilities. Standard errors are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Appendix C Determinants, heterogeneity, and consequence tests

This appendix reports supplementary determinants, heterogeneity, and ancillary consequence tests referenced in the main text. Tables C1, C2, and C3 report determinant specifications; Table C4 reports size heterogeneity in market outcomes; Tables C5–C7 report supporting market and matching checks; and Figure C1 reports the supporting post-ChatGPT event-study path.

Table C1: Reduced-baseline determinants of PatentMismatch

Variable	Dependent variable: any PatentMismatch incident, 2016–2025					
	Log assets	Cash/assets	Leverage	CAPX/assets	ROA	Multivar.
	(1)	(2)	(3)	(4)	(5)	(6)
Log assets	0.012*** (0.004)					0.012*** (0.004)
Cash/assets		-0.024 (0.034)				0.033 (0.033)
Leverage			0.041 (0.036)			0.028 (0.037)
CAPX/assets				0.016 (0.263)		-0.008 (0.281)
ROA					0.013 (0.009)	0.001 (0.009)
Industry FE	Y	Y	Y	Y	Y	Y
All baseline covars.	N	N	N	N	N	Y
Baseline year	2016	2016	2016	2016	2016	2016
Adj. R^2	0.287	0.283	0.283	0.285	0.285	0.287
Observations	2,680	2,680	2,666	2,666	2,676	2,652

Note. This table revisits the firm-level PatentMismatch determinants specification using the reduced baseline characteristic set. The dependent variable is an indicator for whether the firm records at least one PatentMismatch incident during the period 2016–2025. Baseline characteristics are measured in 2016 and include log assets, cash/assets, leverage, CAPX/assets, and ROA. This reduced specification preserves sample size relative to the fuller baseline set. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Table C2: Full-baseline determinants of PatentMismatch

Variable	Dependent variable: any PatentMismatch incident, 2016–2025							
	Log assets	Cash/assets	Leverage	R&D/assets	CAPX/assets	ROA	Employees (k)	Multivar.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Log assets	0.012*** (0.004)							0.009** (0.003)
Cash/assets		-0.024 (0.034)						0.037 (0.046)
Leverage			0.041 (0.036)					0.028 (0.059)
R&D/assets				-0.029 (0.033)				-0.025 (0.050)
CAPX/assets					0.016 (0.263)			-0.497 (0.376)
ROA						0.013 (0.009)		-0.001 (0.017)
Employees (k)							0.000 (0.000)	0.000 (0.000)
Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
All baseline covars.	N	N	N	N	N	N	N	Y
Baseline year	2016	2016	2016	2016	2016	2016	2016	2016
Adj. R^2	0.287	0.283	0.283	0.132	0.285	0.285	0.278	0.133
Observations	2,680	2,680	2,666	1,466	2,666	2,676	2,479	1,331

Note. This table presents the fuller cross-sectional determinants specification for whether a firm records at least one PatentMismatch incident during the period 2016–2025. Baseline characteristics are measured in 2016 and include log assets, cash/assets, leverage, R&D/assets, CAPX/assets, ROA, and the number of employees. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Table C3: Full-baseline determinants of PatentMismatch intensity

Variable	Dependent variable: PatentMismatch share among AI-talking years, 2016–2025							
	Log assets	Cash/assets	Leverage	R&D/assets	CAPX/assets	ROA	Employees (k)	Multivar.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Log assets	0.004 (0.004)							-0.006 (0.005)
Cash/assets		-0.073* (0.042)						-0.045 (0.052)
Leverage			0.060 (0.039)					0.042 (0.073)
R&D/assets				-0.042* (0.023)				-0.078 (0.050)
CAPX/assets					-0.031 (0.165)			-0.434* (0.250)
ROA						0.011* (0.007)		0.002 (0.013)
Employees (k)							0.000 (0.000)	-0.000 (0.000)
Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
All baseline covars.	N	N	N	N	N	N	N	Y
Baseline year	2016	2016	2016	2016	2016	2016	2016	2016
Adj. R^2	0.271	0.272	0.271	0.141	0.272	0.272	0.265	0.135
Observations	2,680	2,680	2,666	1,466	2,666	2,676	2,479	1,331

Note. This companion table uses the share of AI-talking years flagged as PatentMismatch during the period 2016–2025 as the dependent variable rather than the extensive-margin indicator. The baseline characteristics are measured in 2016 and correspond to the full baseline set. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Table C4: Size heterogeneity in filing-date and post-filing mismatch effects

	Small subsample	Big subsample	Pooled interaction
<i>Panel A. Dependent variable: CAR[-1,+1]</i>			
PatentMismatch	0.0013 (0.0065)	0.0014 (0.0044)	0.0011 (0.0041)
PatentMismatch $\times$ Small			0.0033 (0.0059)
Implied mismatch effect, small firms			0.0043 (0.0055)
Small			-0.0036 (0.0043)
AS ratio	-0.0039 (0.0046)	0.0050 (0.0040)	0.0018 (0.0030)
AI Focus	0.0032 (0.0033)	-0.0042 (0.0026)	-0.0010 (0.0020)
Controls	Y	Y	Y
Industry FE	Y	Y	Y
Filing-year FE	Y	Y	Y
Observations	3,556	3,571	7,127
Adj. R^2	0.024	0.003	0.013
<i>Panel B. Dependent variable: BHAR[+2,+63]</i>			
PatentMismatch	0.0102 (0.0231)	-0.0055 (0.0133)	-0.0049 (0.0123)
PatentMismatch $\times$ Small			0.0171 (0.0191)
Implied mismatch effect, small firms			0.0122 (0.0189)
Small			-0.0188 (0.0121)
AS ratio	0.0049 (0.0166)	0.0066 (0.0090)	0.0059 (0.0086)
AI Focus	-0.0037 (0.0114)	-0.0053 (0.0064)	-0.0042 (0.0059)
Controls	Y	Y	Y
Industry FE	Y	Y	Y
Filing-year FE	Y	Y	Y
Observations	3,435	3,473	6,908
Adj. R^2	0.024	0.033	0.025

Note. This table reports heterogeneity in the filing-date and post-filing pricing effects by firm size. Small firms are defined using the monthly active-sample median lagged market capitalization from CRSP. The table is included as supporting evidence because the cleaner return and valuation patterns are concentrated outside the largest firms.

Table C5: Market reactions in capital-raising windows

Panel	Outcome	PM $\times$ Issue	LowCred. $\times$ Issue	AppMismatch $\times$ Issue	Baseline PM	IssueWindow	Mean	Obs.
<i>Panel A. Filing-year market reactions in all issue-eligible firms</i>								
	CAR[-1,+1]	0.0058 (0.0089)	0.0074 (0.0085)	0.0046 (0.0088)	-0.0009 (0.0039)	-0.0055 (0.0055)	0.000	4,378
	BHAR[+2,+63]	0.0716** (0.0346)	0.0577* (0.0313)	0.0526 (0.0341)	-0.0272*** (0.0095)	0.0279 (0.0174)	-0.029	4,376
	BHAR[+2,+252]	0.3885 (0.2433)	0.2496 (0.1922)	0.3498 (0.2353)	0.0094 (0.0242)	0.1729*** (0.0548)	-0.024	4,360
<i>Panel B. Filing-year market reactions in non-big firms</i>								
	CAR[-1,+1]	0.0090 (0.0136)	0.0163 (0.0134)	0.0095 (0.0134)	-0.0014 (0.0080)	-0.0054 (0.0086)	-0.008	2,024
	BHAR[+2,+63]	0.0918* (0.0504)	0.1139** (0.0456)	0.0823* (0.0487)	-0.0353** (0.0176)	0.0282 (0.0241)	-0.068	2,023
	BHAR[+2,+252]	0.4976 (0.3488)	0.3805 (0.2888)	0.4766 (0.3324)	0.0398 (0.0427)	0.1883** (0.0747)	-0.119	2,009

Note. This table reports filing-year return regressions that interact low-credibility AI disclosure measures with the capital-raising window used in Table 7. IssueWindow equals one when next-year CRSP shares outstanding growth exceeds 5%. The three interaction columns report coefficients for the interaction between the listed disclosure measure and IssueWindow. Baseline PM is the PatentMismatch coefficient outside issue windows, and IssueWindow is the level effect in the canonical PatentMismatch specification. Return outcomes are in decimal units. All regressions include filing-year fixed effects, annual controls, and firm-clustered standard errors. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Table C6: Predictive post-filing return regressions with richer controls

	Core controls (1)	+ operating controls (2)	+ market characteristics (3)	+ market + patent history (4)
<i>Panel A. Dependent variable: BHAR[+2,+63]</i>				
PatentMismatch	0.0005 (0.0099)	0.0058 (0.0124)	0.0086 (0.0110)	0.0065 (0.0117)
AS ratio	0.0021 (0.0073)	0.0076 (0.0088)	0.0115 (0.0079)	0.0104 (0.0081)
AI Focus	-0.0044 (0.0053)	-0.0050 (0.0059)	-0.0072 (0.0056)	-0.0063 (0.0059)
Outcome mean	-0.0445	-0.0448	-0.0474	-0.0474
Observations	6,958	4,243	4,113	4,113
Adj. R^2	0.043	0.064	0.227	0.227
<i>Panel B. Dependent variable: BHAR[+2,+252]</i>				
PatentMismatch	0.0216 (0.0266)	0.0206 (0.0319)	0.0322 (0.0224)	0.0365 (0.0234)
AS ratio	-0.0226 (0.0183)	-0.0065 (0.0213)	0.0090 (0.0154)	0.0115 (0.0158)
AI Focus	0.0151 (0.0136)	0.0167 (0.0154)	0.0012 (0.0106)	-0.0007 (0.0108)
Outcome mean	-0.0807	-0.0569	-0.0569	-0.0569
Observations	4,644	2,936	2,936	2,936
Adj. R^2	0.114	0.156	0.536	0.536

Note. This table asks whether post-filing return relations are absorbed by richer observable controls. The unit is an AI-talking annual filing event. Core controls include log assets, leverage, cash/assets, and ROA. Operating controls add R&D/assets, CAPX/assets, and sales growth. Market characteristics add lagged market capitalization, annual return, annual buy-and-hold abnormal return, volatility, and market age. Patent history adds lagged AI patenting. All specifications include SIC2 and filing-year fixed effects, and standard errors are clustered by firm.

Table C7: Matched AI-talking comparison sample

Outcome	Treated mean	Control mean	Difference	p -value	Matched pairs
CAR[-1,+1]	0.0016	0.0006	0.0010	0.812	1,294
BHAR[+2,+21]	-0.0307	-0.0258	-0.0049	0.484	1,270
BHAR[+2,+63]	-0.0513	-0.0477	-0.0036	0.755	1,213
BHAR[+2,+126]	-0.0715	-0.0814	0.0099	0.564	1,121
BHAR[+2,+252]	-0.0648	-0.0840	0.0191	0.523	692

Note. This table compares PatentMismatch filing events with matched AI-talking filing events that are not flagged as PatentMismatch. AI-talking events are filings with at least one AI-related sentence. Matches are exact on filing year and SIC2 industry and nearest on lagged market capitalization, lagged annual BHAR, lagged annual return, log assets, leverage, cash/assets, and ROA, subject to a standardized-distance caliper of 1.50. The matched sample contains 1,297 pairs, with a median match distance of 0.857 and a maximum absolute matched standardized mean difference of 0.025 across the matching covariates. Return outcomes are in decimal units.

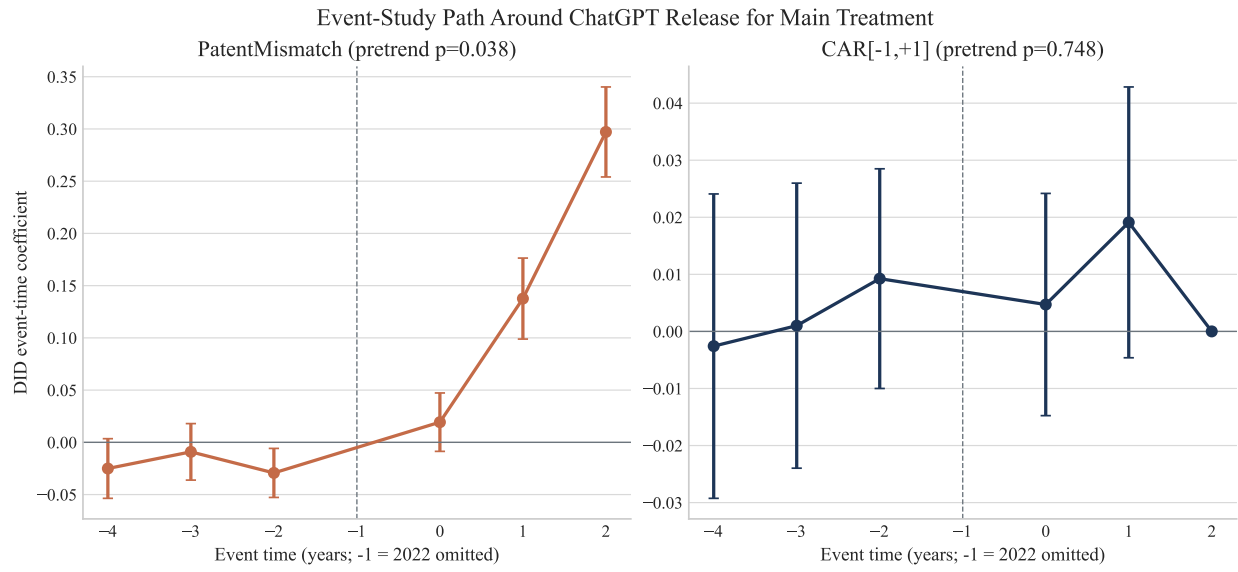


Figure C1: Event-study path around the release of ChatGPT

Note. This figure plots event-time coefficients for the baseline post-ChatGPT treatment around late 2022. The left panel reports PatentMismatch and the right panel reports filing-date $CAR[-1, +1]$. The figure is included as supporting evidence for the post-2022 disclosure-composition patterns and reports pretrend diagnostics to aid interpretation.

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